Suppose I built a correlation matrix for 10 work fields and I want to take only top 5 for each work field and also put an integer rank to it (1 for most similar and 5 for least similar out of top 5).

Now suppose this is my matrix

	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10
A1	1.0	0.9	0.8	0.3	0.7	0.2	0.4	0.5	0.6	0.1
A2	0.9	1.0	0.85	0.2	0.65	0.3	0.5	0.6	0.55	0.15
A3	0.8	0.85	1.0	0.25	0.6	0.35	0.45	0.7	0.5	0.2
A4	0.3	0.2	0.25	1.0	0.4	0.8	0.75	0.35	0.3	0.6
A5	0.7	0.65	0.6	0.4	1.0	0.45	0.5	0.55	0.35	0.25
A6	0.2	0.3	0.35	0.8	0.45	1.0	0.85	0.4	0.3	0.65
A7	0.4	0.5	0.45	0.75	0.5	0.85	1.0	0.55	0.4	0.7
A8	0.5	0.6	0.7	0.35	0.55	0.4	0.55	1.0	0.6	0.3
A9	0.6	0.55	0.5	0.3	0.35	0.3	0.4	0.6	1.0	0.2
A10	0.1	0.15	0.2	0.6	0.25	0.65	0.7	0.3	0.2	1.0

Now you compute the Top 5 most similar attributes for suppose A1 and A5.

You ignore self-correlation (which is 1.0 on the diagonal).

Example:

For A1, the sorted similarities are:

A2 (0.9)
A3 (0.8)
A5 (0.7)
A9 (0.6)
A8 (0.5)

So the Top 5 for A1 are:

A2, A3, A5, A9, A8 (A1 - A5 rank is 3)

Now for A5, the sorted similarities are:

A1 (0.7)
A2 (0.65)
A3 (0.6)
A8 (0.55)
A7 (0.5)

So the Top 5 for A5 are:

A1, A2, A3, A8, A7 (A5 - A1 rank is 1)

Means Correlation (A1, A5) = Correlation (A5, A1) = 0.7, and is symmetrical but

SortedRank(A1, A5) != SortedRank (A5, A1), and it is asymmetrical

So, means, If I remove the duplicates as described in constraint 1. (Symmetry, storing only upper or lower triangle), I won't be getting something like the following:

```
[
  {
        "code1" : "A1"
        "code2" : "A5"
        "value" : 3
        "score" : 0.7
  },
  {
        "code1" : "A5"
        "code2" : "A1"
        "value" : 1
        "score" : 0.7

  }
]
```

Which I think should be there, in order to derive a meaningful correlation matrix.

To summarize: I think removing duplicates is causing the main problem in order to get meaningful correlation matrix and because of this proper candidate matching won't happen as described in the Background and Objective.